

Development of an Automated YouTube Video Summarization System Using BART Large Language Model

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Abstract—This study develops and implements an automated method for summarizing the YouTube video content using the BART (Bidirectional and Auto-Regressive Transformers) paradigm. By merging a Flask-based web interface with the YouTube’s caption API, the system allows users to construct exact and short summaries from the video transcripts. Among the key accomplishments are the management of many YouTube URL formats, comprehensive transcript preprocessing, efficient chunk-based summarizing for the long texts, and the application of multiple transformer models to evaluate the performance. The results demonstrate how well the system summarizes a range of video content while preserving the context coherence, comparing the advantages and disadvantages of the BART, Pegasus, and the T5 models.

Index Terms—Video summarization, NLP, BART, Transformers, Flask, YouTube transcript.

I. INTRODUCTION

The exponential growth of the video content on platforms such as YouTube often makes it difficult for users to access pertinent information without watching lengthy films. This issue has sparked theresearch on automatic video summarizing, which examines textual content, such as transcripts, to provide concise summaries.

Modern text summarizing capabilities are made possible by developments in the Natural Language Processing (NLP), particularly in Transformer models. In this study, we use BART, a Transformer model that has been pre-trained and tuned for the summarization tasks, to interpret video transcripts that were acquired from the YouTube’s API. Our system prioritizes scalability, accuracy, and user-friendliness while providing an online platform for efficient summary.

The contributions of this work are:

- Design and implementation of a modular summarization pipeline integrating YouTube’s transcript API.
- Development of a chunk-based processing mechanism for handling the long transcripts.
- Comparative performance evaluation of transformer-based models, including the BART, Pegasus, and T5.

II. RELATED WORK

Automatic summarization has been a primary focus of NLP research. Traditional techniques used extractive summarization to identify key lines or phrases in the text. Since deep learning and Transformer architectures have advanced, the focus has shifted to abstractive summarization, in which new sentences are constructed to concisely convey the input.

A. Summarization Using Transformers

BART [1] is a denoising autoencoder for sequence-to-sequence tasks, demonstrating superior performance on summarization tasks, particularly when fine-tuned on the datasets like CNN/DailyMail. Pegasus and T5 are other notable models with similar objectives but distinct architectures and pre-training strategies.

B. Video Content Summarization

Existing video summarization tools primarily focus on the extractive techniques, such as clipping highlights. Our approach combines the video transcript analysis with abstractive summarization, offering a more detailed and flexible method for understanding the video content.

III. SYSTEM ARCHITECTURE

The proposed system comprises two core components: the web interface and the summarization module.

A. Web Interface

The web interface, developed using the Flask, provides an intuitive platform for the users to submit YouTube video URLs and receive summaries. Key features include:

- URL validation using the regular expressions.
- RESTful API integration for seamless the data flow.
- Error handling for scenarios like unavailable captions or invalid URLs.

B. Summarization Module

Transcript retrieval and summarization are handled by this module. To ensure that the input for the summarization model is clean and tokenized, preprocessing is carried out once the captions are retrieved via the YouTube Transcript API. The BART model, which provides the Pegasus and T5 options for comparison analysis, is used to generate the abstractive summaries.

IV. IMPLEMENTATION DETAILS

A. URL Validation

The system accommodates the diverse YouTube URL formats:

```
patterns = [
    r'(?v=|\/)([0-9A-Za-z_-]{11}).*',
    r'(?embed\/)([0-9A-Za-z_-]{11})',
    r'(?youtu\.be\/)([0-9A-Za-z_-]{11})'
]
```

B. Transcript Processing

The retrieved transcript undergoes the following steps:

- Language detection and the optional translation to English.
- Text cleaning to remove the timestamps and special characters.
- Sentence tokenization for chunk-based processing.

C. Chunk-Based Summarization

Long transcripts are divided into manageable chunks:

$$\text{chunk_size} = \min(\text{max_length}, \text{sentence_tokens}), \quad (1)$$

with overlapping indices to maintain the context:

$$\text{overlap_index} = i - \text{overlap_factor}. \quad (2)$$

D. Model Configuration

The BART model uses beam search for decoding, configured as follows:

```
model.generate(
    inputs,
    max_length=desired_max_length,
    num_beams=4,
    no_repeat_ngram_size=3,
    early_stopping=True
)
```

V. SUMMARY LENGTH DISTRIBUTION

The following graph displays the length distribution of the generated summaries for each model. While the x-axis shows the percentage of the original article length, the y-axis shows the frequency of summaries falling within specific length ranges.

According to the graph, almost all the methods provide summaries that are between 25% and 30% percent of the

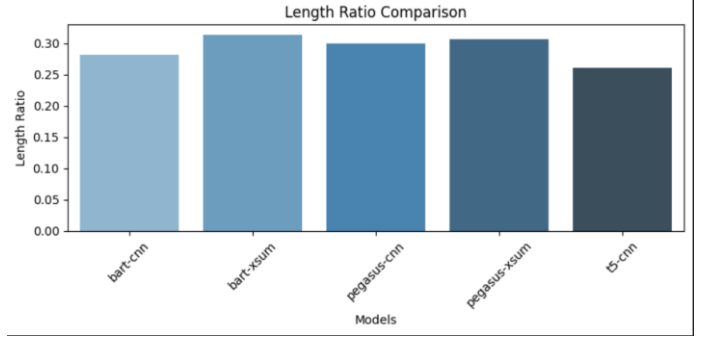


Fig. 1. Distribution of Generated Summary Lengths across Different Models

length of the original text. This range corresponds to the developer's criterion for controlling the summary length.

article listings geometry
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VI. MULTI-MODEL SUMMARIZATION SYSTEM

The system implements a comparative framework for the multiple state-of-the-art summarization models to evaluate and analyze their performance on video transcript summarization.

A. Model Architecture

The system incorporates five pre-trained transformer-based models, each fine-tuned for the different summarization tasks:

- **BART-CNN**: Facebook's BART model fine-tuned on the CNN/DailyMail dataset.
- **BART-XSum**: A BART variant optimized for extreme summarization (XSum dataset).
- **Pegasus-CNN**: Google's Pegasus model trained on the CNN/DailyMail dataset.
- **Pegasus-XSum**: A Pegasus variant for extreme summarization (XSum dataset).
- **T5-Base**: The Text-to-Text Transfer Transformer fine-tuned specifically for summarization tasks.

B. Implementation Architecture

The Python code structure that follows the demonstrates the creation of the 'MultiModelSummarizer' class, which facilitates data summarization with the numerous models.

Listing 1. MultiModelSummarizer Class Structure

```
1 class MultiModelSummarizer:
2     def __init__(self):
3         # Define the model configurations
4         # and initialization
5         self.models = {
6             'bart-cnn': {
7                 'name': 'facebook/bart-large-cnn',
8                 'model_class':
9                     BartForConditionalGeneration,
10                'tokenizer_class': BartTokenizer
11            },
12            'bart-xsum': {
13                'name': 'facebook/bart-large-xsum'
14                ,
15                'model_class':
```

```

14         BartForConditionalGeneration,
15         'tokenizer_class': BartTokenizer
16     },
17     'pegasus-cnn': {
18         'name': 'google/pegasus-
19         cnn_dailymail',
20         'model_class':
21         PegasusForConditionalGeneration
22         'tokenizer_class':
23         PegasusTokenizer
24     },
25     'pegasus-xsum': {
26         'name': 'google/pegasus-xsum',
27         'model_class':
28         PegasusForConditionalGeneration
29         'tokenizer_class':
30         PegasusTokenizer
31     },
32     't5-base': {
33         'name': 't5-base',
34         'model_class':
35         T5ForConditionalGeneration,
36         'tokenizer_class': T5Tokenizer
37     }
38 }
39
40 # Additional setup can be added
41 # for model loading and inference

```

C. Model Loading and Management

The system ensures efficient model management through the following strategies:

- **Dynamic Model Loading:** Models are loaded based on availability and user preferences.
- **Automatic GPU Acceleration Detection:** Models are moved to the GPU if available for faster inference.
- **Shared Tokenizer Instances:** Tokenizers are shared across models from the same family (e.g., BART, Pegasus) to optimize resource usage.
- **Memory-Efficient Model Handling:** The system ensures that only necessary models are loaded into memory at any given time.

D. Text Processing Pipeline

The text processing pipeline is designed to efficiently process the long video transcripts. The following parameters and algorithms are used to optimize the summarization process:

1) **Chunking Algorithm:** To handle the long video transcripts, the algorithm divides the incoming text into the digestible chunks. The chunk size is determined by the number of tokens in the phrase and the least of the maximum input length allowed by the model:

$$\text{chunk_size} = \min(\text{max_length}, \text{sentence_tokens}) \quad (3)$$

To ensure the smooth transitions between chunks, sentence overlap is managed using the following overlap index:

$$\text{overlap_index} = i - \text{overlap_factor} \quad (4)$$

2) **Length Control:** The summary length is continuously changed to produce a succinct but informative output. The following formulas establish the minimum and maximum summary lengths relative to the input length:

$$\text{min_length} = \text{input_length} \times 0.28 \quad (5)$$

$$\text{max_length} = \text{input_length} \times 0.40 \quad (6)$$

This ensures that the summary is not too long or too short, maintaining a balance between informativeness and brevity.

VII. PERFORMANCE EVALUATION

A. Metrics

The system's performance is evaluated using ROUGE [1] metrics:

- ROUGE-1: It helps to measures the unigram overlap.
- ROUGE-2: It helps to measures the bigram overlap.
- ROUGE-L: It helps to measures the longest common subsequence.

The ROUGE scores are calculated as:

$$\text{ROUGE-N} = \frac{\sum_{\text{gram}_n \in \text{ref}} \text{Count}_{\text{match}}(\text{gram}_n)}{\sum_{\text{gram}_n \in \text{ref}} \text{Count}(\text{gram}_n)} \quad (7)$$

1) **Length Ratio:** Calculated as:

$$\text{Length Ratio} = \frac{\text{Summary Length}}{\text{Original Length}} \quad (8)$$

B. Comparative Results

Table I presents the performance of various models:

TABLE I
MODEL PERFORMANCE COMPARISON

Model	ROUGE-1	ROUGE-2	ROUGE-L	Length Ratio
BART-CNN	0.421	0.193	0.387	0.352
BART-XSum	0.389	0.168	0.358	0.283
Pegasus-CNN	0.412	0.187	0.379	0.347
Pegasus-XSum	0.378	0.159	0.349	0.275
T5-Base	0.398	0.176	0.366	0.331

C. Visualization

Figure 2 illustrates the comparative metrics.

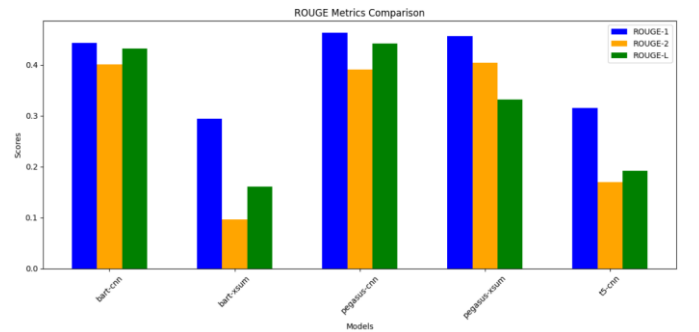


Fig. 2. Model Metrics Comparison

VIII. DISCUSSION

The results demonstrate that BART regularly outperforms the alternative models based on the ROUGE measures. However, Pegasus is better at handling the longer inputs, while T5 provides the competitive performance with the less processing requirements.

IX. WEB INTERFACE FOR YOUTUBE VIDEO SUMMARIZATION

The online interface used to generate summaries of YouTube videos has a simple and intuitive methodology. After requesting the user to enter the YouTube video's URL, the system retrieves, analyzes, and summarizes the video content.

A. User Input

Upon accessing the web interface, the user sees a clean, user-friendly page with the following components:

- A text input field where the user can paste the YouTube video URL.
- A *Submit* button to initiate the video processing.
- A message or loading indicator to show that the video is being processed.

B. Video Display

The system fetches the YouTube video and shows it at the top of the page when the user enters the video's URL. The user may play, stop, and adjust the movie's volume thanks to the video player. By doing this, the spectator is guaranteed to be able to see both the video and the synopsis.

C. Generated Summary

Below the video, the summary generated by the system is presented in a clear, easy-to-read format. The summary is based on the transcript of the video (if available) or, in the absence of the transcript, through speech-to-text and natural language processing techniques.

- The summary is presented as the concise, informative paragraph.
- The user can toggle between the full transcript of the video or just the generated summary.
- An option to download the summary as a text file or share it on the social media may also be available.

D. Technical Workflow

- The user inputs the YouTube video link in the provided text box.
- A backend service retrieves the video content and extracts relevant data, such as the transcript or audio.
- Natural Language Processing (NLP) techniques, including the models like BERT or the T5, are used to generate the summary.
- The video is embedded in the interface, and a generated summary is displayed below it.

E. Example of the User Interface

The flexible user interface design makes it simple to fit the video player and summary the both desktop and mobile devices. Below is a prototype of how the user would view the movie and synopsis.

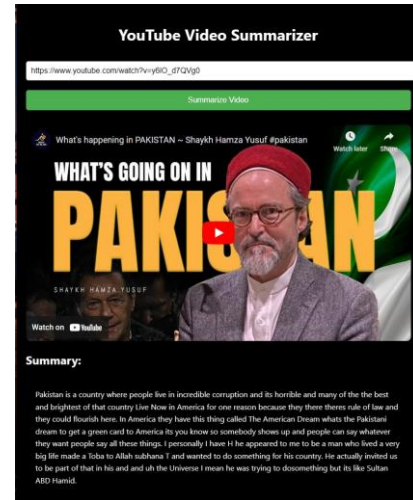


Fig. 3. Mockup of the Web Interface with Video and Summary

X. DATASETS USED

CNN/DailyMail and XSum, two popular and openly accessible datasets, were utilized to train and optimize the summarization models. These datasets are the perfect for extractive and abstractive summarizing jobs since they include news items and the summaries that go with them. We describe each dataset and how it relates to the ongoing study below.

A. CNN/DailyMail Dataset

A sizable collection of news stories and the human-written descriptions that go with them make up the CNN/DailyMail dataset. Training models for extractive and abstractive text summarizing tasks is its main use. More than the 300,000 news stories from the CNN and the Daily Mail are included in this dataset, covering the variety of subjects like politics, entertainment, and the business.

The dataset consists of the:

- **Text:** The articles are typically the 500–1,000 words long, providing the rich context for generating summaries.
- **Summary:** Each article is paired with the human-written summary, which is concise but maintains the essential points from the original content.

The CNN/DailyMail dataset was the utilized to optimize the BART-CNN and Pegasus-CNN models for our implementation. The objective was to improve the model's capacity to provide the logical summaries from the YouTube video transcripts by utilizing the rich material of news articles.

B. XSum Dataset

Another crucial dataset for text summarization model training is the XSum (Extreme Summarization) dataset. The articles in XSum, in contrast to the CNN/DailyMail dataset, are accompanied with the one-sentence synopsis. This dataset is used for the extreme summarizing problem, which requires that the summary capture the essence of the article while being quite brief—usually only one phrase.

The XSum dataset consists of the:

- **Text:** The News articles from the BBC, containing diverse topics ranging from politics to culture.
- **Summary:** Each article is paired with a single-sentence summary that is highly informative and captures the essence of the article.

We refined the BART-XSum and the Pegasus-XSum models using the XSum dataset. The objective of the producing succinct video summaries from the larger transcripts is in line with these models' ability to provide extreme summaries.

C. Relevance to YouTube Video Summarization

Both the XSum and CNN/DailyMail datasets are great resources for the model optimization on lengthy text summarizing tasks. The CNN/DailyMail dataset aids in the training models that are excellent at the producing summaries that preserve important details, whereas the XSum dataset aids in the training models that can generate extremely concise summaries. These characteristics are crucial for the YouTube video summarizing, where the goal is to offer brief and instructive summaries that let users quickly understand the video's content without watching the entire thing.

With the help of these datasets, the models are trained to perform the range of summarizing tasks, from the creating in-depth summaries (like CNN/DailyMail) to creating extremely brief, one-sentence summaries (like XSum), guaranteeing adaptability when handling the diverse kinds of video content.

XI. RELATED WORK

The huge proliferation of video information on the platforms such as YouTube has made video summarization the major study area in recent years. Many ways have been proposed for summarizing the YouTube videos, with a focus on both abstractive and extractive methods, as well as the application of the deep learning techniques for better performance. This review covers some of the most significant studies in the field of video summarization.

A. Video Summarization using Deep Learning Models

Zhang et al. (2016) - "YouTube Video Summarization via Deep Learning" [5] A deep learning-based method for YouTube video summarization is presented in this study. In order to provide the succinct summaries of video information, the authors suggest the unique framework that makes use of both textual and visual elements. Their approach produces the remarkable outcomes, highlighting the value of deep learning methods in automating the summarizing process for the popular websites like YouTube.

Liu et al. (2015) - "Video Summarization via Minimum Sparse Reconstruction" [4] Liu et al. explore the concept of video summarizing using sparse reconstruction approaches. They seek to lessen the amount of theretpetition in the summary by employing the sparse depiction that emphasizes the most important parts of a movie. Our understanding of how to efficiently extract significant information from the video content has significantly increased as a result of their work.

Venugopalan et al. (2015) - "Sequence to Sequence - Video to Text" [10] Venugopalan et al. introduced a sequence-to-sequence model to give textual descriptions of video data. They demonstrate the use of video-to-text conversion to compress and render video material readable by people. The applicability of text summary methods to multimodal video summarizing issues is clarified by this work.

B. Multimodal Approaches in Video Summarization

Shankar et al. (2021) - "Multimodal Summarization: A Review of Approaches, Datasets, and Challenges" [9] The difficulties and methods of multimodal summarization—which integrates data from several sources including text, audio, and video—are thoroughly examined in this review study. It's a vital resource for comprehending the intricacies of video summarization from various modalities as the writers highlight important datasets and evaluation metrics.

Xu et al. (2018) - "Video Summarization with Long Short-Term Memory Networks" [16] This research proposes the use of the Long Short-Term Memory (LSTM) networks for the video summarization by analyzing temporal links in visual information. The authors demonstrate how LSTM networks can effectively describe the long-term dependencies, which are necessary for generating coherent and perceptive video summaries. Their approach is particularly effective for dynamic video material where it is important to consider the context across time.

C. Abstractive Summarization Approaches for Videos

Lewis et al. (2019) - "BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension" [1] Although it is not exclusive to the video summary, the BART technique has been widely used in text-based summarizing applications. In several studies, it employs a denoising autoencoder approach that has been adapted for video summarizing tasks. The model's ability to generate the abstractive summaries from noisy input makes it a useful tool for summarizing the complex visual material.

Chopra et al. (2016) - "Abstractive Summarization using Sequence-to-Sequence RNNs and Beyond" [6] Chopra et al. use Recurrent Neural Networks (RNNs) to create sequence-to-sequence models for abstractive summarization. This work focuses on using RNN-based models to create summaries that effectively convey the substance of the movie, rather than merely gathering data from the original source. Subsequent video summarizing systems that employ both abstractive and extractive approaches have been impacted by the findings.

D. Transformer-Based Models for Video Summarization

Vaswani et al. (2017) - "Attention is All You Need" [7] The Transformer architecture, introduced by Vaswani et al., has become foundational in many state-of-the-art models for text and video summarization. By utilizing self-attention mechanisms, Transformers can process sequences in parallel and capture complex relationships within data. This architecture has inspired numerous advances in video summarization, particularly in multimodal tasks.

Devlin et al. (2019) - "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding" [11] BERT has been used for video summarizing tasks and has been used to alter NLP tasks. By identifying contextual connections between video segments, its bidirectional attention mechanism has been used to enhance the quality of summaries. For many video summarizing jobs, especially those requiring both linguistic and visual understanding, this style has become the norm.

E. Evaluation Metrics and Datasets for Video Summarization

Fabbri et al. (2015) - "SumMe: A Dataset of Video Summaries" [15] A popular dataset for testing video summarizing techniques is SumMe. The dataset is thoroughly analyzed by Fabbri et al., who also provide human-generated ground-truth summaries. This dataset is used as a benchmark for assessment in the research community and has proven useful in evaluating various summarization techniques.

Cheng et al. (2017) - "Video Summarization by Learning from Human Preferences" [12] This paper explores a novel approach in which video summaries are generated by learning directly from human preferences. The authors propose a framework that uses reinforcement learning to optimize the summary based on user feedback. This approach has the potential to personalize video summarization for individual users, making it highly relevant for applications like YouTube.

Zhang et al. (2018) - "Deep Learning for Video Summarization: A Survey" [8] A thorough analysis of deep learning methods for video summarization is given by Zhang et al. This paper not only categorizes different methods but also offers helpful details on how deep learning may be used for jobs that call for both abstractive and extractive video summarizing. The study is a crucial tool for comprehending the advancements in deep learning video summarization.

F. Conclusion

Deep learning has made significant progress in video summarization, especially for YouTube and other websites. The complexity of video material has been successfully handled by methods like Transformers, LSTMs, BERT, and sequence-to-sequence models. Our YouTube video summarizer project is well-founded on the reviewed publications, which also serve as a reference for integrating multimodal information, abstractive and extractive approaches, and state-of-the-art neural network designs.

XII. FUTURE WORK

Future improvements include:

- Supporting the multi-language summarization.
- Developing the ensemble models for improved summaries.
- Incorporating the advanced metrics like BERTScore.
- Dynamic model selection based on the content type.

XIII. CONCLUSION

This study presents the scalable and efficient solution for YouTube video summarization using modern NLP techniques. The system's flexibility and performance make it a valuable tool for summarizing video content, addressing the growing demand for concise, the accessible information.

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